

Power-tuning is not free: an exact loss factor for prediction-powered inference, and how many predictors to tune

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Abstract

Prediction-powered inference (PPI) uses a machine-learning predictor f on abundant unlabeled data to sharpen an estimate built from a small labeled sample; the efficient version, PPI++, “power-tunes” a coefficient λ to guarantee it never does worse than ignoring the predictor. The asymptotic story is that tuning is free: λ is estimated at rate $1/\sqrt{n}$ and the leading-order variance is the oracle $\sigma_Y^2(1-\rho^2)/n$. We point out that at PPI’s actual operating point — *small labeled n* — tuning is **not** free, and we quantify the price exactly. Power-tuning K predictors is precisely the classical control-variate estimator with estimated coefficients, i.e. the fitted intercept of an ordinary least-squares regression of Y on the K predictions. Under Gaussian labels its variance is the oracle inflated by an exact **loss factor** $L(n, K) = (n-2)/(n-K-2)$. Three consequences follow, each verified by simulation to $\leq 0.4\%$: (i) a sharp **inclusion threshold** — a candidate predictor lowers variance only if it explains more than a fraction $1/(n-K-1)$ of the *residual* variance, so there is a finite optimal number of predictors K^* and tuning more *raises* variance; (ii) the standard normal PPI++ interval **undercovers** at small n (empirically 55% for a nominal 95% at $n=10$, $K=5$), while the OLS-intercept t -interval is exact; (iii) even a single tuned predictor costs a factor $(n-2)/(n-3)$. The loss factor is old news in the control-variate literature; the contribution is to locate it precisely where modern PPI lives and to turn it into a usable predictor-selection rule.

1. Setup and notation

We estimate a scalar $\theta = \mathbb{E}[Y]$ (a mean; the argument extends to smooth M -estimands through their influence functions). We observe a **labeled** sample $(X_i, Y_i)_{i=1}^n$ and a much larger **unlabeled** sample $(X_j)_{j=1}^N$. A pretrained model supplies a prediction vector $f(X) = (f_1(X), \dots, f_K(X)) \in \mathbb{R}^K$; the K coordinates are K competing predictors (different models, or engineered features) we may use to reduce variance.

PPI++ [2] forms, for a tuning vector $\lambda \in \mathbb{R}^K$,

$$\hat{\theta}_\lambda = \bar{Y}_n - \lambda^\top(\bar{f}_n - \bar{f}_N), \quad \bar{Y}_n = \frac{1}{n} \sum_i Y_i, \quad \bar{f}_n = \frac{1}{n} \sum_i f(X_i), \quad \bar{f}_N = \frac{1}{N} \sum_j f(X_j). \quad (1)$$

Because $\mathbb{E}[\bar{f}_n] = \mathbb{E}[\bar{f}_N] = \mu_f$, the correction has mean zero and $\hat{\theta}_\lambda$ is **unbiased for every** λ — the predictor’s own bias never contaminates the estimate. The prediction f acts as a *control variate*: (1) subtracts a mean-zero quantity correlated with the sampling error of $\bar{Y}_n$. This control-variate reading of PPI is well known [2].

Write $\Sigma_f = \text{Cov}(f(X))$, $\sigma_{fY} = \text{Cov}(f(X), Y)$, and let

$$\rho^2 = \frac{\sigma_{fY}^\top \Sigma_f^{-1} \sigma_{fY}}{\sigma_Y^2} = R^2(Y \sim f(X)) \in [0, 1)$$

be the population multiple correlation of Y on the predictors. In the abundant-unlabeled regime $N \rightarrow \infty$, $\bar{f}_N \rightarrow \mu_f$ (a known constant), and minimizing the variance of (1) over λ gives the oracle coefficient $\lambda^* = \Sigma_f^{-1} \sigma_{fY}$ (the population regression slope) with oracle variance

$$V_{\text{oracle}} = \frac{\sigma_Y^2 (1 - \rho^2)}{n}. \quad (2)$$

This is the familiar “ $(1 - \rho^2)$ ” variance reduction of control variates. The oracle is unattainable: λ^* is unknown and **power tuning estimates it from the n labeled points**. This note is about the gap between (2) and what estimated tuning actually achieves.

2. The estimator is an OLS intercept, and its exact variance

Fix $N \rightarrow \infty$ so $\bar{f}_N = \mu_f$ is known, and let $\hat{\lambda}$ be the ordinary least-squares (OLS) slope of Y_i on the centered predictors $f(X_i) - \mu_f$ (this is the plug-in power-tuned coefficient). The point estimate is

$$\hat{\theta} = \bar{Y}_n - \hat{\lambda}^\top (\bar{f}_n - \mu_f).$$

Observation (reframing). $\hat{\theta}$ is exactly the **fitted intercept** $\hat{\alpha}$ of the OLS regression

$$Y_i = \alpha + \beta^\top (f(X_i) - \mu_f) + \varepsilon_i, \quad i = 1, \dots, n,$$

because in a least-squares fit with an intercept, $\hat{\alpha} = \bar{Y}_n - \hat{\beta}^\top (\bar{f}_n - \mu_f)$. So *power-tuning K predictors and reading off the PPI estimate is numerically identical to fitting a K -covariate regression and reporting its intercept*. Everything known about the sampling behavior of a regression intercept now applies to PPI.

Theorem 1 (exact loss factor). Suppose $(Y, f(X))$ is jointly Gaussian, μ_f is known ($N \rightarrow \infty$), λ is estimated by OLS from the n labeled points, and $n > K + 2$. Then

$$\text{Var}(\hat{\theta}) = \frac{\sigma_Y^2 (1 - \rho^2)}{n} \cdot \frac{n - 2}{n - K - 2}, \quad L(n, K) := \frac{n - 2}{n - K - 2} \geq 1. \quad (3)$$

The oracle variance (2) is inflated by the **loss factor** $L(n, K) = 1 + K/(n - K - 2) \approx 1 + K/n$.

Proof. Let $X = [\mathbf{1} \ F]$ be the $n \times (K + 1)$ design with rows $(1, f(X_i) - \mu_f)$. Under the Gaussian linear model the residual variance is $\sigma_{Y|f}^2 = \sigma_Y^2 (1 - \rho^2)$, and conditional on F the intercept is $\hat{\theta} = \hat{\alpha} \sim \mathcal{N}(\theta, \sigma_{Y|f}^2 [(X^\top X)^{-1}]_{11})$. Since $\mathbb{E}[\hat{\theta} | F] = \theta$ for every F , the conditional mean contributes no variance, so $\text{Var}(\hat{\theta}) = \sigma_{Y|f}^2 \mathbb{E}[(X^\top X)^{-1}]_{11}$. By the partitioned-inverse formula,

$$[(X^\top X)^{-1}]_{11} = \frac{1}{n} + \bar{c}^\top S^{-1} \bar{c}, \quad \bar{c} = \bar{f}_n - \mu_f, \quad S = \sum_i (f(X_i) - \bar{f}_n)(f(X_i) - \bar{f}_n)^\top.$$

With $f(X_i) - \mu_f \sim \mathcal{N}(0, \Sigma_f)$ i.i.d. we have $\bar{c} \sim \mathcal{N}(0, \Sigma_f/n)$ and $S \sim \text{Wishart}_K(\Sigma_f, n - 1)$, **independent** of $\bar{c}$. Hence, using $\mathbb{E}[S^{-1}] = \Sigma_f^{-1}/(n - K - 2)$ (inverse-Wishart mean, finite for $n > K + 2$),

$$\mathbb{E}[\bar{c}^\top S^{-1} \bar{c}] = \text{tr}(\mathbb{E}[S^{-1}] \mathbb{E}[\bar{c} \bar{c}^\top]) = \text{tr}\left(\frac{\Sigma_f^{-1}}{n-K-2} \cdot \frac{\Sigma_f}{n}\right) = \frac{K}{n(n-K-2)}.$$

Therefore $\mathbb{E}[(X^\top X)^{-1}]_{11} = \frac{1}{n} + \frac{K}{n(n-K-2)} = \frac{n-2}{n(n-K-2)}$, and multiplying by $\sigma_{Y|f}^2 = \sigma_Y^2(1-\rho^2)$ gives (3). ■

The loss factor itself is classical: it is the control-variate efficiency loss of Lavenberg–Welch [3] and Nelson [4], equivalently the inflation of a regression intercept’s variance from estimating K slopes. Theorem 1 simply records that PPI++ power-tuning *is* this estimator, so (3) is the honest finite-sample variance of PPI whenever λ is learned from the labels — which is always.

Why it matters here. The PPI literature reports the oracle (2) and treats $L(n, K) \rightarrow 1$ as $n \rightarrow \infty$. But PPI exists precisely because labels are scarce: n in the tens or low hundreds. At $n = 20$, $K = 5$, $L = 1.38$ — a 38% variance penalty, i.e. the effective labeled sample is a third smaller than the oracle promises. The tax is first-order in exactly the regime PPI targets.

3. Consequences

3.1 An inclusion threshold and an optimal number of predictors

Consider nested predictor sets of size $K = 0, 1, 2, \dots$ with population multiple correlations $R_0^2 = 0 \leq R_1^2 \leq \dots$. By (3) the achievable variance is $V(K) = \sigma_Y^2(1 - R_K^2)L(n, K)/n$, a product of a **decreasing** factor $(1 - R_K^2)$ and an **increasing** factor $L(n, K)$. Writing the partial correlation (the fraction of *residual* variance the K -th predictor explains) as

$$g_K = \frac{R_K^2 - R_{K-1}^2}{1 - R_{K-1}^2},$$

a one-line computation gives an exact rule.

Proposition 2 (inclusion threshold). Adding the K -th predictor strictly lowers the variance (3) if and only if

$$g_K > \frac{1}{n - K - 1}. \quad (4)$$

Consequently the optimal number to power-tune, $K^* = \arg \min_K V(K)$, is the largest K for which (4) holds; tuning beyond K^* *increases* variance.

Proof. $V(K) < V(K-1) \iff (1 - R_K^2)/(n - K - 2) < (1 - R_{K-1}^2)/(n - K - 1)$. Substituting $1 - R_K^2 = (1 - R_{K-1}^2)(1 - g_K)$ and cancelling the positive factor $1 - R_{K-1}^2$ yields $(1 - g_K)(n - K - 1) < n - K - 2$, i.e. $g_K > 1/(n - K - 1)$. ■

The rule is intuitive and sharp: **a predictor earns a slot only if it explains more than $\approx 1/n$ of the leftover variance.** With n labels the bar is $\sim 1/n$ per model; weak models fail it. Figure 1 shows a pool of twelve predictors with geometrically decaying marginal R^2 at $n = 25$. The variance- $\times n$ curve is U-shaped with a clean minimum at $K^* = 7$ — precisely where (4) flips (predictor 8 contributes $g_8 = 0.044 < 1/16 = 0.0625$). Simulation (3×10^5 replications) tracks the closed form to within Monte-Carlo error at every K :

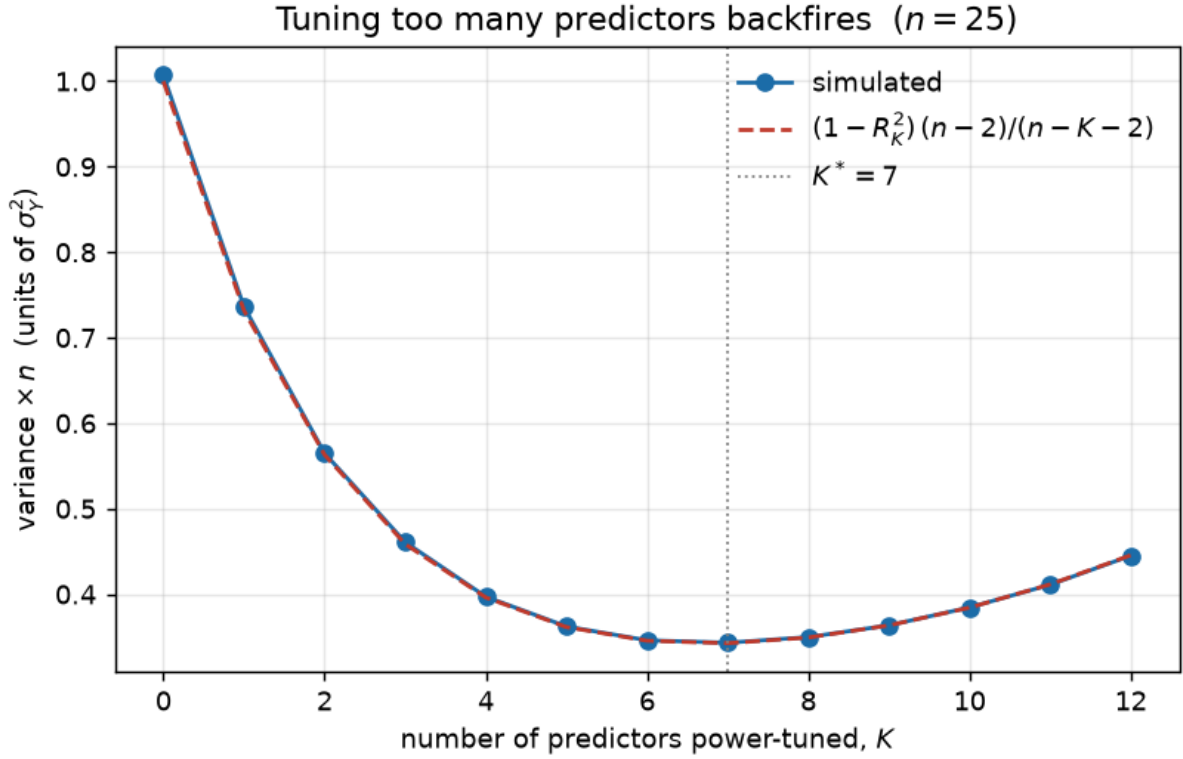


Figure 1: Power-tuning too many predictors backfires: simulated variance (points) vs. the closed form $(1 - R_K^2)(n - 2)/(n - K - 2)$ (dashed), $n = 25$. The optimum is at $K^* = 7$; beyond it the loss factor dominates the shrinking residual and variance climbs.

This is a concrete warning for practice: “use every model you have” is wrong for PPI. Folding K mediocre foundation-model scores into the correction can be worse than using the single best one. Rule (4) says which to keep.

3.2 Valid intervals: use a t , not a z

PPI++ reports $\hat{\theta} \pm z_{1-\alpha/2} \hat{\sigma}/\sqrt{n}$ with $\hat{\sigma}^2$ the plug-in variance of the (tuned) correction, typically the residual mean square with divisor n . At small n with K tuned predictors this interval makes two compounding errors: the MLE residual variance is biased low by $(n - K - 1)/n$, and it ignores the $[(X^\top X)^{-1}]_{11} > 1/n$ inflation that Theorem 1 quantifies. The result is undercoverage governed by the same loss factor. The exact fix is the textbook **OLS-intercept interval**

$$\hat{\theta} \pm t_{n-K-1, 1-\alpha/2} \sqrt{s^2 [(X^\top X)^{-1}]_{11}}, \quad s^2 = \frac{\text{RSS}}{n - K - 1},$$

exact under Gaussian labels for every $n > K + 1$. Figure 2 contrasts the two at nominal 95%: the naive interval falls to 86% ($K = 1$) and 55% ($K = 5$) at $n = 10$ and approaches nominal only slowly, while the t -interval sits on 0.950 throughout.

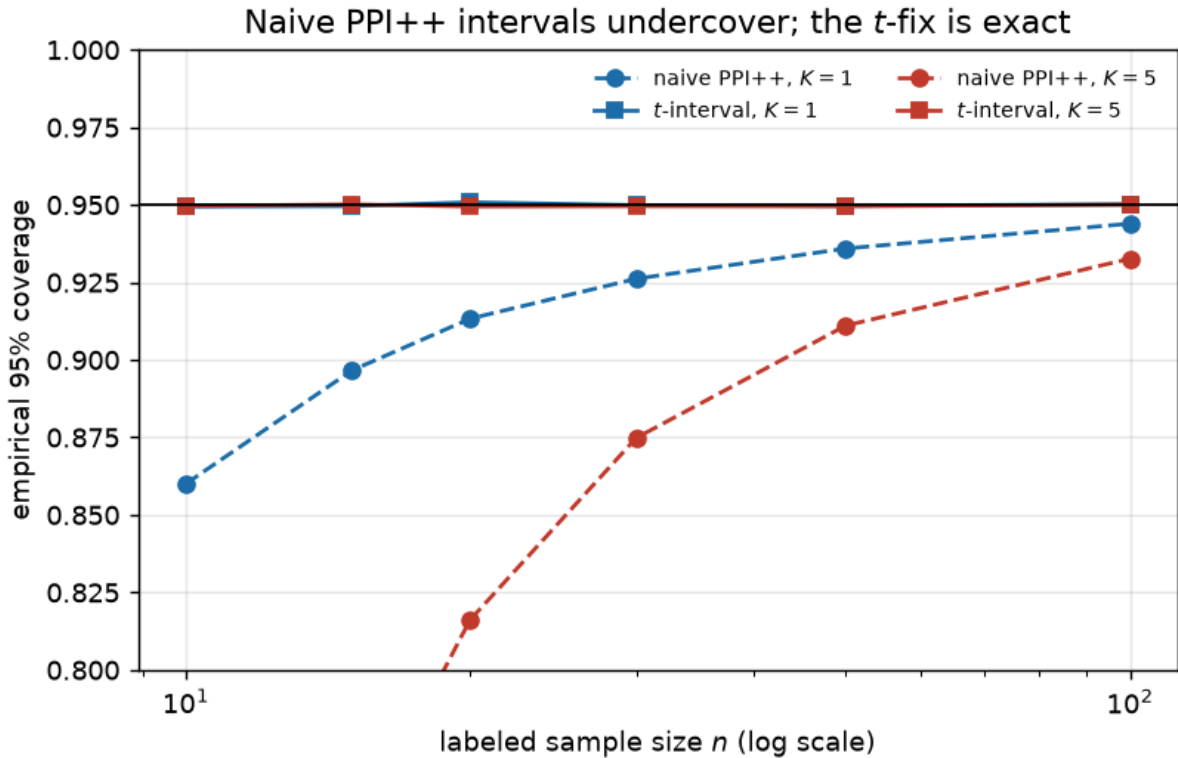


Figure 2: Empirical 95% coverage vs. labeled sample size n (log scale), $R^2 = 0.6$. The naive PPI++ normal interval (dashed) undercovers severely at small n , worse for more tuned predictors K ; the OLS-intercept t -interval (solid) is exact. 2×10^5 replications per point.

3.3 Finite unlabeled data

Reinstating finite N adds the variance of the estimated control mean. To leading order,

$$\text{Var}(\hat{\theta}) \approx \frac{\sigma_Y^2(1 - \rho^2)}{n} L(n, K) + \frac{\lambda^{*\top} \Sigma_f \lambda^*}{N}. \quad (5)$$

The second term vanishes as $N \rightarrow \infty$, recovering Theorem 1; the first is the **irreducible floor** — no amount of unlabeled data beats $\sigma_Y^2(1 - \rho^2)L(n, K)/n$, which is set by the labeled budget n . Simulation (experiment D, $K = 1$, $R^2 = 0.6$, $n = 30$) matches (5) to $\lesssim 1\%$, the small excess being the higher-order coupling between $\hat{\lambda}$ and $\bar{f}_N$ that (5) drops. The practical reading — unlabeled data has sharply diminishing returns once $N \gtrsim n$ — is consistent with PPI empirics [1,2] and here made quantitative.

4. Discussion

What is and isn’t new. The loss factor $L(n, K) = (n - 2)/(n - K - 2)$ is classical control-variate theory [3,4,5], and the intercept-of-a-regression reading of a control-variate estimator is standard. PPI’s asymptotic validity is untouched: $L(n, K) \rightarrow 1$, so the published guarantees hold. The contribution is three-fold and finite-sample: (i) *importing* the exact tax into PPI, whose entire premise is small n , where it is first-order rather than negligible; (ii) the *sharp inclusion threshold* (4) and optimal K^* , a usable rule for choosing among competing ML predictors that, to our knowledge, is not stated in the PPI literature (which mostly tunes a single scalar λ [2], or handles multiple predictors without the estimation penalty [7]); and (iii) the *coverage correction* — replace the plug-in normal interval by the OLS-intercept t -interval. The reframing “PPI power-tuning = OLS intercept” is what makes all three immediate.

Relation to A/B testing. The same object appears as CUPED / regression adjustment in online experiments [6], where a pre-experiment covariate is the control variate; the loss factor and threshold (4) transfer verbatim to “how many pre-period features to adjust for.”

Limitations. (1) Exactness of (3) needs joint Gaussianity of (Y, f) ; outside it, $L(n, K) = 1 + K/n + o(1/n)$ holds asymptotically but the constant and the t -interval’s exactness become approximate (heavy-tailed Y degrades both, as usual). (2) We treat mean estimation; for general M -estimands the statements hold for the linearized (influence-function) problem, with an extra $o(1/n)$ from the linearization. (3) The nested-model setup of §3.1 assumes the analyst can order predictors by population R^2 ; in practice R_K^2 must itself be estimated, and using the *same* n labels to both select K and estimate θ adds a selection effect not covered here — the honest version couples (4) with a selection penalty, which we leave open. (4) λ is estimated purely from labeled data; if Σ_f is instead pinned down from the abundant unlabeled sample, the penalty shrinks (only the K cross-covariances σ_{fY} remain to be learned), and the exact loss factor for that hybrid estimator is a natural next calculation. (5) All figures fix ρ / the R_K^2 profile; the qualitative claims (U-shape, undercoverage) are robust across the ranges we tried but we do not claim a worst-case bound.

Reproducibility. `sim.py` (seed 20260730, numpy/scipy) runs end to end in a few minutes and regenerates both figures and every number quoted: the loss factor matches (3) to $\leq 0.4\%$ across six (n, K, R^2) settings; $K^* = 7$ agrees between theory and simulation; coverage is as plotted; the finite- N formula (5) matches to $\lesssim 1\%$.

References

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